# WORD EMBEDDING

## Problem 1.

(5 points) Prove that softmax is invariant to constant offsets in the input, that is, for any input vector $\boldsymbol{x}$ and any constant $c$,

$$\text{softmax}(\boldsymbol{x}) = \text{softmax}(\boldsymbol{x} + c)$$

where $\boldsymbol{x} + c$ means adding the constant $c$ to every dimension of $\boldsymbol{x}$. Remember that

$$softmax(\boldsymbol{x})_i = \frac{e^{\boldsymbol{x}_i}}{\sum_j e^{\boldsymbol{x}_j}} \tag{1}$$

*Note: In practice, we make use of this property and choose* $c = -\max_i x_i$ *when computing softmax probabilities for numerical stability (i.e., subtracting its maximum element from all elements of* $\boldsymbol{x}$*).*

## Problem 2.

(3 points) Derive the gradients of the sigmoid function and show that it can be rewritten as a function of the function value (i.e., in some expression where only $\sigma(x)$, but not $x$, is present). Assume that the input $x$ is a scalar for this question. Recall, the sigmoid function is

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

(3 points) Derive the gradient with regard to the inputs of a softmax function when cross entropy loss is used for evaluation, i.e., find the gradients with respect to the softmax input vector $\boldsymbol{\theta}$, when the prediction is made by $\hat{\boldsymbol{y}} = \text{softmax}(\boldsymbol{\theta})$. Remember the cross entropy function is

$$CE(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -\sum_i y_i \log(\hat{y}_i) \tag{3}$$

where $\boldsymbol{y}$ is the one-hot label vector, and $\hat{\boldsymbol{y}}$ is the predicted probability vector for all classes. (*Hint: you might want to consider the fact many elements of* $\boldsymbol{y}$ *are zeros, and assume that only the* $k$*-th dimension of* $\boldsymbol{y}$ *is one.*)